Food Delivery App for Thailand

1. Cost of Launching :

- Google Play Store : $25 USD / 839 THB, One Time Payment to permanently launch
- Apple App Store : $99 USD / 3825 THB, Yearly Renewal Required
- For buying website : $12.34 / year

2. Points of Importance :

1. Cost doesn't depend on the user count, it highly depends on the number of orders as well.
2. We can use Free quotas / Pay less until we've high users & orders, when we've high traffic then we will need Amazon Web Services (AWS) for maintenance which is expensive.
3. More Users -> Higher database storage + Higher Menu Image downloads + Higher Mapbox Map loads (So more expensive)

3. Phases of the App :

1. Phase 1 : Minimum Viable Product (MVP)

- We will mostly utilise the free services provided here, apart from server management which depends on the number of users.
- Approx Cost for
    1. 1k - 5k Active Users / Month is : ₹4.2k - ₹5k p.m
        - Main Cost Drivers : Basic Server and Database Cost [Render Pro + Supabase Pro]
    2. 5k – 10k Active Users / Month is : ₹13k - ₹17k p.m

- Main Cost Drivers : Same As Last One + Mapbox [Mapbox is basically used for Resturants seeing when the driver arrives + Used by the driver for finding route to restaurant & delivery location + To show the predicted delivery time when someone opens their cart + Searching for an address + Showing Nearby Resturants + Used by customer to check delivery driver's location]

2. Phase 2 : Growth

1. 1k - 5k Active Users / month and 5k – 20k orders/month
   - Approx Cost : ₹20k – ₹24k p.m
   - Main Cost : Sending OTPs [Exceeds free quota we get /month] + Phone Number Authentication Required + More Database Expansion Cost + More Server Cost from Render + Redis Cost
2. 5k – 10k Active Users / month and 10k – 20k orders/month
   - Approx Cost : ₹45k – ₹55k p.m
   - Main Cost : Better backend servers + Mapbox takes around 35k of the bill + Supabase Cost
3. 5k – 10k Active Users / month and 20k – 25k orders/month
   - Approx Cost : ₹75k - ₹85k p.m
   - Main Cost : Websockets + Routing APIs + Mapbox
4. 10k – 25k Active Users / month and 20k – 25k orders/month
   - Approx Cost : ₹75k - ₹85k p.m
   - Main Cost : Delivery Tracking (websockets) + Driver Routing API Calls +
   - Main Changes : Change Mapping -> OSRM to reduce costs.

3. Phase 3 : High Density

1. 25k – 50k Active Users / month and 50k – 75k orders/month
   - Approx Cost : ₹2.5 Lacs - ₹3 Lacs
   - Main Cost : Must migrate to AWS services now so cost increase a lot here.
   - Managing such high traffics is incredibly tough, AWS automates these services and therefore higher expenditure is required.

- Required Technologies : AWS ECS (Fargate) + AWS RDS PostgreSQL Multi-AZ + Redis + OSRM

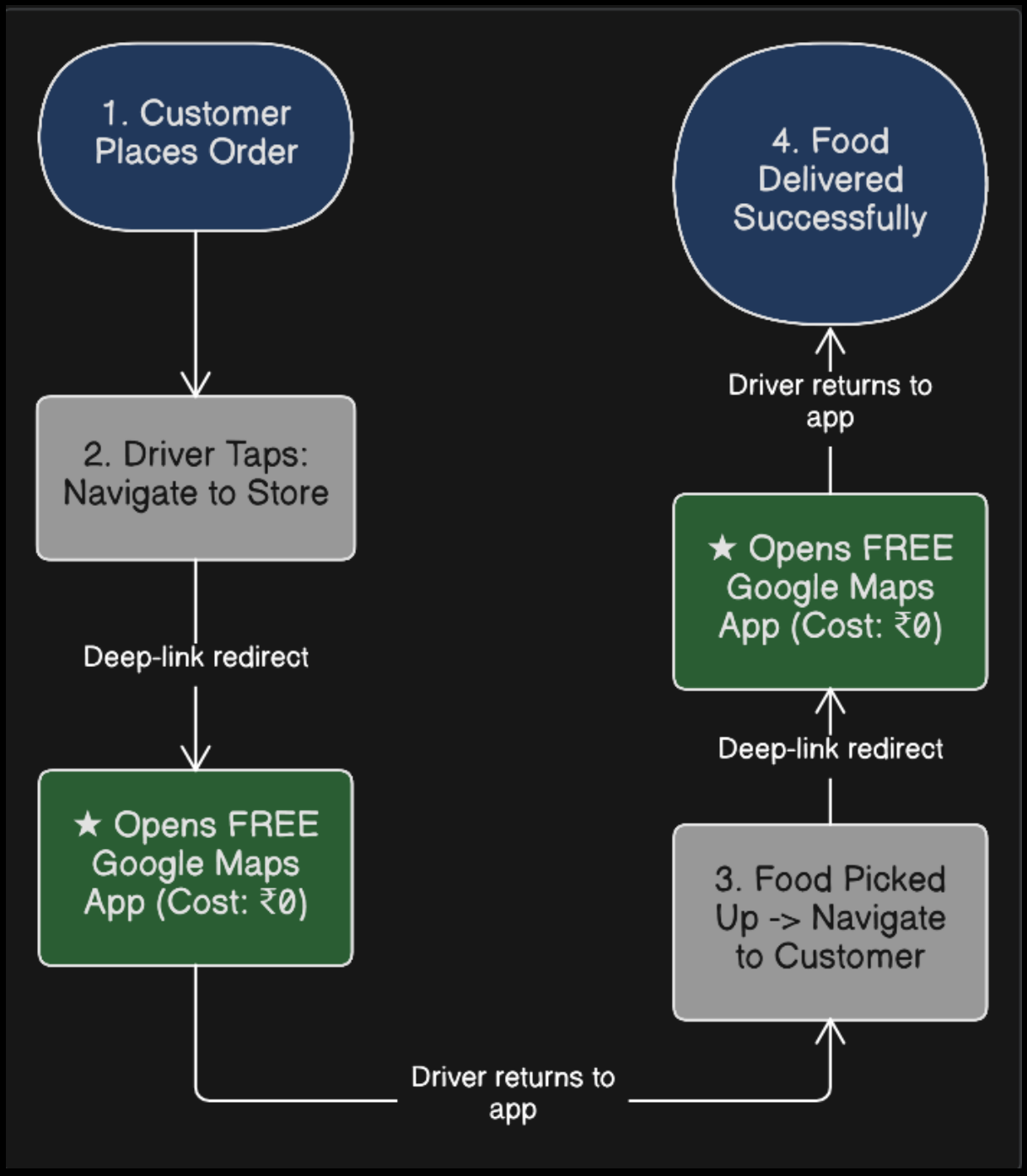
4. Phase 4 : Enterprise

1. 50k – 100k Active Users / month and 100k – 150k orders/month
    - Mapbox alone costs ₹3 Lacs+
    - SMS OTPs cost ₹80k+
    - Required Technologies : AWS EKS (Kubernetes) + Aurora Postgres + Self-Hosted OSRM + Kafka

4. Saving Costs

- If we use an option using which the Driver clicks it and then gets redirected to Google Maps for the route from his location to the restaurant, and then later clicks it to find the route from the restaurant

to the delivery location, then we save the Mapbox high API costs a lot which are incurred in this



- So the new costs are :

Phase & Scale Tier	Old Cost / month	New Cost / month	Money Saved
Phase 1 (Tier 1.1): 1k–5k MAU (0–1k Orders)	₹4,200 – ₹5,000	₹4,200 – ₹5,000	₹0 (Inside Free Tier)
Phase 1 (Tier 1.2): 5k–10k MAU (0–1k Orders)	₹13,000 – ₹17,000	₹12,000 – ₹15,000	₹1,000 – ₹2,000
Phase 2 (Tier 2.1): 1k–5k MAU (5k–20k Orders)	₹20,000 – ₹24,000	₹14,000 – ₹18,000	~₹6,000
Phase 2 (Tier 2.2): 5k–10k MAU (10k–20k Orders)	₹45,000 – ₹55,000	₹32,000 – ₹40,000	₹13,000 – ₹15,000
Phase 2 (Tier 2.3): 5k–10k MAU (20k–25k Orders)	₹75,000 – ₹85,000	₹50,000 – ₹60,000	~₹25,000
Phase 2 (Tier 2.4): 10k–25k MAU (20k–25k Orders)	₹75,000 – ₹85,000	₹55,000 – ₹65,000	~₹20,000
Phase 3 (Tier 3.1): 25k–50k MAU (50k–75k Orders)	₹2.5 Lacs – ₹3.0 Lacs	₹1.8 Lacs – ₹2.2 Lacs	₹70,000 – ₹80,000
Phase 4 (Tier 4.1): 50k–100k MAU (100k–150k Orders)	₹4.8 Lacs – ₹5.5 Lacs	₹3.8 Lacs – ₹4.5 Lacs	~₹1.0 Lac+

5. Payment Costs :

- Integration cost of Stripe (or Omise), portal required for payment into our app is free.
- After that the cost stands as :

Payment Method	Base Gateway Fee	7% VAT (Value Added Tax) on Fee	Total Effective Fee (TEF)
PromptPay (QR Code)	1.65%	+0.12%	**1.77%**
Credit / Debit Cards (Visa, MC, JCB)	3.65%	+0.26%	**3.91%**
Mobile Banking / E-Wallets (TrueMoney)	1.65% – 1.75%	+0.12%	**1.87%**
International Credit Cards (Amex, Foreign MC)	4.50% – 5.00%	+0.35%	**5.35%**

6. Costs Not Included For :

- Customer Service
- Password Resetting + Restaurant daily summaries
- Payment Gateway Transaction Fees
- SMS Pumping and Toll Fraud Protection Services
- Legal Fees to setup Thai Company + Registration fees for **ETDA** (Electronic Transactions Development Agency) + Compliance costs for **PDPC** (Personal Data Protection Act - Thailand's GDPR equivalent).
- Direct payouts to delivery partners